Reg. No										

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.Sc DEGREE EXAMINATION APRIL 2021

ASMT202T - MATHEMATICAL STATISTICS - II

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2 = 20)

Answer ALL the questions

1. Define Binomial distribution.
2. State Geometric distribution?
3. Define Uniform distribution?
4. Define Exponential distribution.
5. State Degrees of Freedom.
6. Define - null hypothesis.
7. Write Type I and Type II error.
8. State Parameter and statistic.
9. State the principle of ANOVA technique.
10. Define Randomization.

SECTION-B (5×5=25)

Answer ALL the questions

11. (a) Write Mean and Variance of Poisson distribution.

(Or)

(b) Explain MGF of mean and variance of Geometric distribution.

12. (a) Write any five properties of Normal distribution.

(Or)

(b) Prove that limiting form of t - distribution.

13. (a) Explain test of independent of attributes.

(Or)

(b) Write F – test for equality of two population variance.

14. (a) The life time of electric bulbs a random sample of 10 from a large consignment gave the following data.

Line in hours : 4.2, 4.6, 3.9, 4.1, 5.2, 3.8, 3.9, 4.4, 5.6

Can we accept the hypothesis that the average life time is 4,000 hours.

(Or)

(b) Explain test of significance for single Proportion.

15. (a)Write the procedure for analysis of one way ANOVA.

(Or)

(b) Write the Assumption of ANOVA test.

SECTION – C (3 × 10 = 30)

Answer any **THREE** Question

16. Explain Recurrence relation for the moments of Binomial distribution.
17. Derive χ^2 - distribution.
18. Before an increase in excise duty on tea, 400 people out of a sample of 500 people were found to be tea – drinkers. After an increase in duty 400 people were tea- drinkers in a sample of 600 people, state whether there is any significant decrease in the consumption of tea.
19. Explain the test for the significance of the difference of proportions.
20. Analyse the following Randomized Block Design.

Blocks				
Treatments	I	II	III	IV
I	38	46	36	39
II	42	40	38	36
III	45	43	42	44
IV	40	45	45	46

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